MA1521 L1 Week 11 Mon 20261026

Supply limits for iterated integrals in spherical coordinates $\iiint d\rho\, d\phi\, d\theta$ for each of the following regions. (No integrand is specified; $d\rho\, d\phi\, d\theta$ is given so as to determine the order of integration.)

a) The region bounded below by the cone $z^2 = x^2 + y^2$, and above by the sphere of radius $\sqrt{2}$ and center at the origin.

b) The first octant.

c) That part of the sphere of radius 1 and center at $z = 1$ on the z-axis which lies above the plane $z = 1$.

Find the center of mass of a hemisphere of radius a, using spherical coordinates. Assume the density $\delta = 1$.

Evaluate: $\iiint_W (x^2 + y^2 + z^2)^{-1/2}\, dx\, dy\, dz$ where W is the region determined by $\frac{1}{2} \leq z \leq 1$ and $x^2 + y^2 + z^2 \leq 1$.

(a previous question, but now use spherical coordinates instead)

Evaluate: $\iiint_W \exp(x^2 + y^2 + z^2)^{3/2}\, dV$ where W is the unit ball in $\mathbb{R}^3$.